Embedded Systems Lab

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Task 1

Traffic Light System

- First scenario of a traffic light system
  - One line and one traffic light (red, yellow, green light)
  - Control the lights with respect to timing
    - Use time events like t20 for the presentation of the switching delay
- Model the traffic light system with the following diagrams
  - Structure: a class diagram that represents the main structural elements
  - State machine diagram for each class
    - What are the possible states, transitions, events, …?
    - Test all relevant scenarios!
  - Use for a fast “drawing” the open draw.io tool
- Develop a simulation for the traffic light control system
  - Make yourself familiar with Arduino simulators like Tinkercad Circuits or Wokwi and realize a simulation for your traffic light realization
  - Use leds as traffic light
  - Use an Arduino as µC
  - Map the state machine model to code (via switch state pattern!)
  - Develop a circuit for leds
    - Determine the required voltage, current and resistant of the series resistor
    - For the circuit design you can use MultiSim
- Create a private git repository (at github) based on your university mail account
  - Upload all relevant models/documents to the repository - > professionally push updates of for your models/documents to your repository!
- Deadline:
  - Submit solution to git latest October 6 eob